

PAVAN DHARMA ADAPA

Software Engineer · Data & Analytics

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PROFESSIONAL SUMMARY

Software and data engineer with an M.S. in Computer Science (Artificial Intelligence and Robotics). Works the full width of a data problem — pipeline, tests, packaging and analysis — with 34 merged pull requests to an open-source scientific tool and public analyses on real CMS, CDC and de-identified hospital data, negative results included. Work-authorized through June 2027 on F-1 OPT and eligible for the 24-month STEM extension: no sponsorship required until 2029. Available immediately.

TECHNICAL SKILLS

Languages & Tools: Python, SQL, Bash, R, Git and GitHub, Linux, macOS, WSL2, Jupyter, Advanced Excel

Software Engineering: Continuous integration (GitHub Actions matrices), pytest and test-suite design, static portability analysis, packaging (pip, conda), cross-platform debugging, code review via fork-and-PR, technical documentation

Data, Statistics & ML: ETL/ELT, workflow orchestration (Snakemake), dimensional modeling and star schemas, SQLite, data quality gates, scikit-learn, A/B testing and CUPED, power analysis, calibration

PROFESSIONAL EXPERIENCE

Data Analytics Intern | *Agricultural Microbiomes Lab, Mississippi State University, Starkville, MS* Jul 2026 – Present

- Primary contributor to **nano16s**, the lab's open-source Nanopore 16S rRNA profiling pipeline (Snakemake, Python, Bash) — **34 merged pull requests, 53 of 64 non-merge commits and +5,700 lines** under maintainer review.
- Diagnosed and fixed a cross-platform failure blocking **all macOS users** — an unguarded empty-array expansion aborting the classification stage — then added macOS to the weekly CI pipeline, which on its first run surfaced a second platform bug that reported healthy installs as broken.
- Extended the test suite to **87 tests** including a static portability scanner that catches shell-compatibility regressions on every pull request, and maintained a CI matrix across Linux/macOS and Python 3.10/3.12.

Graduate Teaching Assistant | *Mississippi State University, Starkville, MS* Sep 2025 – May 2026

- Taught data cleaning, transformation and visualization in Python and R to **100+ students across six courses**, and reported completion and grade metrics that let instructors reach at-risk students earlier.

Graduate Research Assistant | *Mississippi State University, Starkville, MS* Aug 2024 – Aug 2025

- Built the automated collection and cleaning workflow that standardized multi-source research data, **cutting per-dataset processing from hours to under 10 minutes** and making every figure reproducible from source.
- Built and compared forecasting models — Random Forest, XGBoost and Gaussian Process Regression — and defined the baseline metrics used to track their performance over time.

SOFTWARE & DATA PROJECTS

Self-directed public projects — code, tests and continuous integration on GitHub; the data source is named on each.

Experimentation Toolkit — *A/B test failure modes · simulation against known ground truth · Python, SciPy*

- Measured five A/B test failure modes against known ground truth rather than asserting them: peeking ten times at a null experiment drives the false-positive rate to **18.5% against a nominal 5%**, and CUPED at rho 0.7 cuts variance **49%**.
- Showed sample-ratio-mismatch detection depends entirely on scale — a 0.4% split imbalance is $p = 3e-141$ at 10M users and $p = 0.26$ at 20k — and validated every sample-size calculation by simulation: asked for 80% power, measured 80.1%.

Clinical Quality Measures Engine — *Python, YAML specs, pytest · Synthea synthetic non-PHI EHR*

- Designed a measures engine over a **13,810-patient EHR** (820,993 encounters, 10.8M observations) where each measure is a versioned YAML spec with its value sets held outside the engine — an annual update is a data diff, not a code change.
- Decomposed a diabetes control measure to show **203 of 204 failures were patients never tested**, reframing it as a testing-outreach problem rather than a clinical one.

EDUCATION

M.S., Computer Science — Mississippi State University, Mississippi State, MS May 2026

B.Tech, Computer Science — Malla Reddy University, Hyderabad, India May 2024

Publication (under revision): Li Zhang, Saikanth Ratnavale, Pavan Dharma Adapa, Sai Harika Gade, Michael E. Navicky — Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production.